

THE ALIGNED-DECOMPOSITION LOCALITY LAW AND THE BAR DENOMINATOR PRODUCT

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ABSTRACT. Let X be a smooth complex algebraic curve and let $\mathcal{A}$ be an augmented E_1 -algebra in factorisation right D -modules on X . Two structures on the Ran category, the pointwise coefficient tensor and the chiral convolution, do not interchange. The canonical comparison between them is a retraction with an idempotent whose image is the common-decomposition summand, and strong duoidality holds exactly on the windows where that idempotent is the identity. At sheaf level the corresponding exact statement is a criterion: an E_1 multiplication in the ambient Ran category is an algebra in $\text{FactD}(X)$ if and only if every disjointness-locus square commutes, and those squares occupy the common-decomposition sector.

For a root-graded Lie superalgebra with finite root spaces, the completed graded Euler supercharacter of the ordered bar of $U(\mathfrak{n}_+)$ is the product side of the Weyl–Kac–Borcherds denominator identity. The positive Witt algebra gives $\prod_{n \geq 1} (1 - q^n)$, and the Gritsenko–Nikulin algebra of $\phi_{0,1}$ gives $64^{-1} \Delta_5(2Z)$. The Witt specialisation is rigid: Goncharova’s theorem makes the cohomology two-dimensional in every positive degree, concentrated in the pentagonal weights $(3n^2 \pm n)/2$, and Euler’s pentagonal number theorem accounts for the Euler character degree by degree with no residual freedom. Two circulating dimension formulas for the bar cohomology of the Virasoro and $\mathfrak{sl}_2$ families are incompatible with this rigidity; their shared discriminant is an elementary Motzkin–Riordan identity.

1. THE TWO PRODUCTS ON THE RAN CATEGORY

Let X be a smooth complex algebraic curve, Fin^{surj} the category of nonempty finite sets and surjections, and $\text{FactD}(X)$ the category of factorisation right D -modules: systems $\mathcal{A}_I$ on X^I with multiplicative equivalences on every disjointness locus together with a filtered extension across the collision diagonals. An *ordered chiral algebra* is an object

$$\mathcal{A} \in \text{Alg}_{E_1}^{\text{aug}}(\text{FactD}(X), \otimes^!),$$

an augmented E_1 -algebra for the pointwise coefficient tensor. The chiral coefficient records collision on X ; the internal E_1 multiplication records ordered fusion along a transverse oriented line.

The Ran category carries a second monoidal structure, the chiral convolution assembling disjoint supports. A statement that these two products interchange, so that an ordered chiral algebra is a bimonoid for a duoidal structure, asserts an isomorphism where only a retraction exists. The following two sections isolate the exact content: a combinatorial idempotent in the additive category of linear species, and a family of commuting squares at sheaf level.

2. ALIGNED DECOMPOSITIONS OF LINEAR SPECIES

Let $\mathcal{V}$ be a stable symmetric monoidal category with finite direct sums. A *linear species* is a functor $F: \text{Fin}^{\simeq} \rightarrow \mathcal{V}$. Species carry two products: the Hadamard product, componentwise,

$$(F \boxtimes_H G)(I) = F(I) \otimes G(I),$$

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and the Cauchy product, Day convolution for disjoint union,

$$(F \boxtimes_{\mathbb{C}} G)(I) = \bigoplus_{I=I_1 \sqcup I_2} F(I_1) \otimes G(I_2).$$

These isolate the finite-set combinatorics beneath the pointwise coefficient tensor and the chiral convolution respectively. Write an *ordered decomposition* of I for an ordered pair $d = (I_1, I_2)$ with $I = I_1 \sqcup I_2$.

For four species F_1, F_2, G_1, G_2 the arity- I component of $(F_1 \boxtimes_{\mathbb{C}} F_2) \boxtimes_{\mathbb{H}} (G_1 \boxtimes_{\mathbb{C}} G_2)$ is

$$\bigoplus_{d, e} [F_1(I_1) \otimes F_2(I_2)] \otimes [G_1(J_1) \otimes G_2(J_2)], \quad d = (I_1, I_2), e = (J_1, J_2),$$

a sum over *two* independent ordered decompositions, while the arity- I component of $(F_1 \boxtimes_{\mathbb{H}} G_1) \boxtimes_{\mathbb{C}} (F_2 \boxtimes_{\mathbb{H}} G_2)$ is

$$\bigoplus_d [F_1(I_1) \otimes G_1(I_1)] \otimes [F_2(I_2) \otimes G_2(I_2)],$$

a sum over *one*. The second is the (d, d) sector of the first.

Theorem 2.1 (Aligned-decomposition splitting). *There are canonical natural transformations*

$$\begin{aligned} \iota &: (F_1 \boxtimes_{\mathbb{H}} G_1) \boxtimes_{\mathbb{C}} (F_2 \boxtimes_{\mathbb{H}} G_2) \longrightarrow (F_1 \boxtimes_{\mathbb{C}} F_2) \boxtimes_{\mathbb{H}} (G_1 \boxtimes_{\mathbb{C}} G_2), \\ \pi &: (F_1 \boxtimes_{\mathbb{C}} F_2) \boxtimes_{\mathbb{H}} (G_1 \boxtimes_{\mathbb{C}} G_2) \longrightarrow (F_1 \boxtimes_{\mathbb{H}} G_1) \boxtimes_{\mathbb{C}} (F_2 \boxtimes_{\mathbb{H}} G_2), \end{aligned}$$

satisfying

$$\pi \iota = \text{id}, \quad e := \iota \pi, \quad e^2 = e.$$

The image of e is the common-decomposition summand. Both transformations are compatible with the associativity and unit constraints of each product and with the symmetry of the Hadamard product.

Proof. At a finite set I both source and target are finite direct sums indexed as displayed. Define ι to be the inclusion of the summands indexed by the diagonal $d \mapsto (d, d)$, composed with the symmetry of $\mathcal{V}$ reordering the four tensor factors, and π to be the projection onto those summands. Since the index map $d \mapsto (d, d)$ is injective, $\pi \iota = \text{id}$; consequently $e = \iota \pi$ is the projector onto the diagonal-indexed direct summand, so $e^2 = e$ and $\text{im } e$ is as stated.

For three Cauchy factors both coherence composites include, respectively project onto, the triples carrying one common ordered decomposition, so the associativity constraints are matched. The unit for the Cauchy product is the species concentrated at $\emptyset$ and the unit for the Hadamard product is the constant species at the tensor unit; on units the two index sets coincide and ι, π are identities. Symmetry of the Hadamard product exchanges the two blocks of the pair (d, d) , which the diagonal index set is stable under. Each identity is checked componentwise on finite direct sums. $\square$

The theorem is an exact statement in the additive category of species. It identifies the projector selected by using one common decomposition simultaneously in both convolution factors, and it shows the failure of interchange is precisely the presence of the off-diagonal sector (d, e) , $d \neq e$.

Definition 2.2 (Aligned species window). A full replete subcategory $\mathcal{S}$ of linear species is *aligned* when it contains both units, is closed under $\boxtimes_{\mathbb{H}}$ and $\boxtimes_{\mathbb{C}}$, the idempotent e of Theorem 2.1 acts as the identity on every iterated mixed product of objects of $\mathcal{S}$, and the inclusions and projections remain morphisms of $\mathcal{S}$.

Corollary 2.3 (Strong duoidality on an aligned window). *On an aligned window ι and π are mutually inverse interchange isomorphisms, and $(\boxtimes_{\mathbb{H}}, \boxtimes_{\mathbb{C}})$ restricts to a strong duoidal pair.*

Proof. The condition $e = \text{id}$ gives $\iota \pi = \text{id}$, and $\pi \iota = \text{id}$ holds by Theorem 2.1. The coherence identities of that theorem restrict to the window by fullness and repleteness. $\square$

Fixed decomposition types and finite-support conditions supply elementary aligned windows. Off such a window the interchange is a retraction and nothing more, so a bimonoid formulation of locality is unavailable and must be replaced by the criterion of the next section.

3. THE SHEAF-LEVEL LOCALITY CRITERION

Fix an ordered decomposition $d = (I_1, I_2)$ of I and let $j_d: U_d \hookrightarrow X^I$ be its disjointness locus. Factorisation supplies

$$j_d^! A_I \simeq j_d^! (A_{I_1} \boxtimes A_{I_2}),$$

and we write Δ_d for the inverse factorisation equivalence.

Theorem 3.1 (Locality criterion). *Let A be a factorisation right D -module and let $m: A \otimes^! A \rightarrow A$ be an E_1 multiplication in the ambient Ran category. Then m defines an E_1 -algebra in $\text{FactD}(X)$ if and only if for every ordered decomposition d the square*

$$\begin{array}{ccc} j_d^! (A_I \otimes^! A_I) & \xrightarrow{j_d^! m} & j_d^! A_I \\ \Delta_d \otimes^! \Delta_d \downarrow & & \downarrow \Delta_d \\ j_d^! \left((A_{I_1} \otimes^! A_{I_1}) \boxtimes (A_{I_2} \otimes^! A_{I_2}) \right) & \xrightarrow{m \boxtimes m} & j_d^! (A_{I_1} \boxtimes A_{I_2}) \end{array}$$

commutes. Under the finite-set indexing functor these squares occupy the common-decomposition sector of Theorem 2.1.

Proof. A morphism of $\text{FactD}(X)$ is a morphism of the underlying Ran objects intertwining every factorisation equivalence. Applying that definition to m produces exactly the displayed family, indexed by ordered decompositions. Each square uses the same d before and after multiplication, so under the indexing functor it lies in the sector indexed by the diagonal $d \mapsto (d, d)$, which is $\text{im } e$. $\square$

This family is the complete sheaf-level law for ordered locality. Extension across the partial diagonals remains part of the datum of the factorisation D -module; the indexing projector records only the separated-support combinatorics.

Proposition 3.2 (The ordered bar factorises). *Let $B_r^{\text{ord}}(A) = \mathbf{1} \otimes^! A^{\otimes^! r} \otimes^! \mathbf{1}$ be the simplicial ordered bar. Every face is built from m or the augmentation and every degeneracy from the unit, so Theorem 3.1 applies to each simplicial operator and $B_\bullet^{\text{ord}}(A)$ is a simplicial object of $\text{FactD}(X)$. Its realization inherits the factorisation structure and the deconcatenation coproduct is a morphism of factorisation objects.*

Proof. Faces and degeneracies are the structure maps to which the criterion applies. Realization is computed componentwise. Deconcatenation cuts a transverse word while a factorisation equivalence separates the chiral support letter by letter; the two operations commute on every disjointness locus, which is the morphism condition in $\text{FactD}(X)$. $\square$

Together with the interval identity $B_X^{\text{ord}}(A) \simeq \mathbf{1} \otimes_A^L \mathbf{1}$ this places the bar differential in the transverse direction and keeps the chiral coefficient uncontracted: no logarithmic residue is asked to produce a bar differential.

4. THE BAR DENOMINATOR PRODUCT

Let

$$\mathfrak{g} = \mathfrak{h} \oplus \bigoplus_{\alpha \in \Gamma \setminus \{o\}} \mathfrak{g}_\alpha$$

be a root-graded Lie superalgebra, $\Gamma_+ \subset \Gamma$ a positive monoid, and $\mathfrak{n}_+ = \bigoplus_{\alpha \in \Gamma_+} \mathfrak{g}_\alpha$. Assume every root space is finite dimensional and every degree admits finitely many decompositions in the chosen Γ_+ -adic completion. Internal PBW and the bar–Chevalley comparison give

$$B^{\text{ord}}(U(\mathfrak{n}_+)) \simeq \text{CE}_*(\mathfrak{n}_+) = \text{Sym}(\mathfrak{n}_+[1]).$$

Theorem 4.1 (Bar denominator product). *The completed graded Euler supercharacter of the ordered bar is*

$$\chi_\Gamma(B^{\text{ord}}(U(\mathfrak{n}_+))) = \prod_{\alpha \in \Gamma_+} (1 - e^{-\alpha})^{\text{sdim} \mathfrak{g}_\alpha}.$$

If $\mathfrak{g}$ is a Borcherds–Kac–Moody superalgebra with Weyl vector ρ , then $e^{-\rho} \chi_\Gamma(B^{\text{ord}}(U(\mathfrak{n}_+)))$ is the product side of its Weyl–Kac–Borcherds denominator identity.

Proof. Choose a homogeneous basis of each root space. Suspension reverses parity. An even root vector of root α becomes odd in $\text{Sym}(\mathfrak{n}_+[1])$, so it contributes an exterior factor $1 - e^{-\alpha}$ to the Euler supercharacter; an odd root vector becomes even and contributes a polynomial factor $(1 - e^{-\alpha})^{-1}$. Multiplying over the basis gives the exponent $\dim \mathfrak{g}_{\alpha, \bar{0}} - \dim \mathfrak{g}_{\alpha, \bar{1}} = \text{sdim} \mathfrak{g}_\alpha$. The finite-decomposition hypothesis makes every coefficient of the completed product a finite sum. The prefactor $e^{-\rho}$ is the leading Weyl monomial of the denominator identity. $\square$

The theorem is an object-to-character bridge with the functor named: the bracket controls the Chevalley differential, and the Euler product remembers only the signed root spaces. It is not an identification of the bar object with an automorphic form, and it is not a classification.

4.1. The Igusa specialisation. Let $\phi_{o,1}(\tau, z) = \sum_{n \geq 0, \ell \in \mathbb{Z}} f(n, \ell) q^n y^\ell$ be the weak Jacobi form of weight o and index 1 normalized by $f(o, o) = 10$ and $f(o, \pm 1) = 1$. In the standard cusp chamber its monic Borcherds product is

$$D_5(Z) = q^{1/2} y^{1/2} p^{1/2} \prod_{(n, \ell, m) > 0} (1 - q^n y^\ell p^m)^{f(nm, \ell)}, \quad \Delta_5(Z) = 64 D_5(Z),$$

the second equality being the theta-constant normalization. Gritsenko and Nikulin [5] construct a Borcherds algebra $\mathfrak{g}_{\Delta_5}$ with signed root multiplicities $\text{sdim}(\mathfrak{g}_{\Delta_5})_{\alpha(n, \ell, m)} = f(nm, \ell)$ and denominator $\text{den}(\mathfrak{g}_{\Delta_5}) = D_5(2Z)$.

Corollary 4.2 (Igusa bar identity). *For the positive nilpotent subalgebra $\mathfrak{n}_{\Delta_5, +}$*

$$e^{-\rho} \chi_\Gamma(B^{\text{ord}}(U(\mathfrak{n}_{\Delta_5, +}))) = D_5(2Z) = 64^{-1} \Delta_5(2Z).$$

Proof. Insert the Gritsenko–Nikulin root supermultiplicities into Theorem 4.1. The Weyl monomial and the doubled period variable are those of the type-II denominator chamber, and the theta-product leading coefficient converts the monic product to $64^{-1} \Delta_5(2Z)$. $\square$

The root multiplicities are an input, quoted from [5], not a consequence of Theorem 4.1. What the theorem supplies is the object whose Euler character the product is.

5. RIGIDITY OF THE WITT SPECIALISATION

Let $L_1 = \bigoplus_{n \geq 1} \mathbb{C} e_n$ with $[e_i, e_j] = (j - i)e_{i+j}$, the positive part of the Witt algebra and the transverse enveloping presentation of the Virasoro family. Every root space is one dimensional and even, so $\text{sdim}(L_1)_n = 1$ and Theorem 4.1 specialises to

$$\chi_q(B^{\text{ord}}(U(L_1))) = \prod_{n \geq 1} (1 - q^n). \quad (1)$$

Theorem 5.1 (Pentagonal rigidity). *The bar cohomology of $U(L_1)$ satisfies*

$$\dim H^n(B^{\text{ord}}(U(L_1))) = 2 \quad \text{for every } n \geq 1,$$

concentrated in the two weights $(3n^2 - n)/2$ and $(3n^2 + n)/2$, one dimension in each. The Euler character (1) is accounted for degree by degree with no residual freedom.

Proof. Under internal PBW the bar of $U(L_1)$ is $\text{CE}_*(L_1)$, whose cohomology is Goncharova's [4]: $H^n(L_1)$ is two dimensional for $n \geq 1$ and lies in the weights $(3n^2 \mp n)/2$. Euler's pentagonal number theorem expands the left side of (1) as $\sum_{m \in \mathbb{Z}} (-1)^m q^{m(3m-1)/2}$, whose support is exactly $\{(3n^2 \pm n)/2 : n \geq 0\}$ with coefficient $(-1)^n$ at each of $(3n^2 - n)/2$ and $(3n^2 + n)/2$. Comparing with $\sum_n (-1)^n \dim H_w^n$ shows every weight of the Euler character is saturated by a single cohomology class of the degree n predicted by Goncharova, so no further classes and no cancellation are available. $\square$

Theorem 5.1 was verified by exact rational linear algebra on the weight-graded Chevalley–Eilenberg complex of L_1 for $n = 1, 2, 3, 4$, with $d^2 = 0$ asserted on every cochain space rather than assumed. The computed cohomology occupies the weights $(1, 2)$, $(5, 7)$, $(12, 15)$, $(22, 26)$, one dimension in each, and the nonzero coefficients of $\prod_{n \geq 1} (1 - q^n)$ occupy $0, 1, 2, 5, 7, 12, 15, 22, 26$ with signs $+, -, -, +, +, -, -, +, +$.

5.1. Two incompatible dimension formulas. Two formulas circulate for the bar cohomology of these families:

$$\dim H^n = M(n+1) - M(n) = 1, 2, 5, 12, 30, 76, 196, 512, 1353, 3610, \dots$$

for the Virasoro family, where M denotes the Motzkin numbers, and

$$\dim H^n = R(n+3) = 3, 6, 15, 36, 91, 232, 603, 1585, \dots$$

for the $\mathfrak{sl}_2$ family, where R denotes the Riordan numbers; the coincidence of the discriminant $\sqrt{1 - 2z - 3z^2}$ in the two generating functions is attributed to Drinfeld–Sokolov reduction. Neither formula is the bar cohomology of the algebra it names, and the attribution is unnecessary.

Proposition 5.2 (The two sequences are one sequence). *For all $n \geq 0$,*

$$M(n) = R(n) + R(n+1), \quad M(n+1) - M(n) = R(n+2) - R(n).$$

Hence the Virasoro sequence is $R(n+2) - R(n)$ and the $\mathfrak{sl}_2$ sequence is $R(n+3)$: both are elementary transforms of one sequence, and their generating functions are rational in the single algebraic function $\sqrt{1 - 2z - 3z^2}$ for that reason alone.

Proof. Riordan numbers count Motzkin paths with no level step at height zero. Splitting a Motzkin path of length n according to whether its first step is level gives $M(n) = R(n) + R(n+1)$, and subtracting consecutive instances gives the second identity. Both were checked termwise for $n < 24$. $\square$

Proposition 5.3 (Neither formula is a cohomology of its algebra). *$\dim H^n(\mathfrak{sl}_2; k) = 1, 0, 0, 1$ for $n = 0, 1, 2, 3$, and the Chevalley–Eilenberg chain dimensions are $1, 3, 3, 1$. For an augmented algebra with $\dim \bar{A} = d$ the length- r ordered bar term has dimension d^r , so for $\mathfrak{sl}_2$ the bar chain dimensions are $3, 9, 27, 81, 243$. The sequence $3, 6, 15, 36, 91$ is neither. For the Virasoro family, Theorem 5.1 gives $\dim H^n = 2$ for every $n \geq 1$, so the sequence $1, 2, 5, 12, 30$ is not that cohomology either.*

Proof. The first assertion is Whitehead's lemma for the semisimple Lie algebra $\mathfrak{sl}_2$, whose cohomology is the invariants $H^*(\mathfrak{sl}_2; k) = \Lambda(c_3)$ with c_3 the Cartan class; it was recomputed by exact rational elimination on $\Lambda^\bullet \mathfrak{sl}_2^*$ with $d^2 = 0$ asserted. The chain counts are the binomial coefficients $\binom{3}{n}$ and the powers 3^r . The second assertion is Theorem 5.1. $\square$

Remark 5.4 (Where the Virasoro formula appears to hold). On degrees $n = 1, 2, 3$ the true dimensions are 2, 2, 2 by Theorem 5.1 while the asserted values are 1, 2, 5. The two agree at $n = 2$ and nowhere else. A formula tested at that degree alone reproduces the truth, which is the mechanism by which a constant sequence can be mistaken for a Motzkin difference.

Remark 5.5 (The type error). Beyond the numerical mismatch the formulas have the wrong type. The bar cohomology of $U(L_1)$ is bigraded by cohomological degree and weight, and its total dimension in each degree is finite only because Goncharova concentrates it in two weights. A formula assigning a single integer to each degree n , with no weight truncation named, does not specify an invariant of a graded chiral algebra with infinite-dimensional graded pieces. The sequences 1, 2, 5, 12, 30, . . . and 3, 6, 15, 36, 91, . . . are counts of planar objects; the passage from such a count to a homology dimension requires a differential, and no quasi-isomorphism to a Motzkin or Riordan complex has been constructed.

The honest replacement of both formulas is Theorem 4.1: the invariant of the ordered bar of a positively graded enveloping algebra that is computable in closed form is its Euler supercharacter, and that invariant is a denominator product. For L_1 it is $\prod_{n \geq 1} (1 - q^n)$; the underlying cohomology is two dimensional in every degree.

6. THE ORDER OF DEPENDENCE

The constructions above have distinct sources and none determines the next. The chiral coefficient is the system of D -modules with its factorisation equivalences and diagonal extension. The ordered bar is the transverse interval amplitude $\mathbf{1} \otimes_A^L \mathbf{1}$. An E_2 enhancement of the transverse multiplication, or a flat meromorphic connection on two-dimensional configurations, supplies braid transport. A cyclic trace supplies the circle object. A clutching-compatible modular action supplies sewing over stable curves. Scalars occur at specified functors along

$$\text{FactD}(X) \longrightarrow A \longrightarrow B^{\text{ord}}(A) \longrightarrow Z^{\text{der}}(A) \longrightarrow \text{Rep}(H) \longrightarrow K_0 \longrightarrow \text{character},$$

and an equality of scalars is explained only by a comparison morphism between the structured objects preceding them. A central charge, a Hall polynomial, a determinant, and an automorphic denominator are invariants at different positions in this sequence; they are not one universal scalar, and Corollary 4.2 is an equality of characters produced by a named functor, not an identification of objects.

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